

ICS Bayesian Risk Assessment Report

Executive Summary

This report presents the results of a quantitative Bayesian risk assessment for an Industrial Control System (ICS) environment. The assessment uses a Bayesian network model to compute compromise probabilities and risk scores based on the system topology, asset attributes, and any observed evidence.

Metric	Value
Overall Risk Score	5.116
Risk Level	Critical
Assets Assessed	11
Connections Assessed	12
Evidence Used	None

Risk Register

Assets ranked by risk score (posterior probability × consequence impact). Higher scores indicate higher priority for investigation or treatment.

Rank	Asset	Risk Score	Probability	Risk Level
1	scada	6.618	0.945	—
2	physical_process	5.216	0.373	—
3	local_hmi	4.787	0.957	—
4	plc_2	4.479	0.383	—
5	plc_1	4.479	0.383	—
6	sensors_actuators_2	3.470	0.334	—
7	sensors_actuators_1	3.470	0.334	—
8	industrial_network	3.078	0.770	—
9	historian	1.185	0.237	—
10	operators	0.000	0.210	—
11	automation_engineers	0.000	0.146	—

Attack Path Analysis

The following attack paths represent the most likely routes an adversary could take through the network to reach high-value targets, ranked by combined propagation-and-target-risk score.

Path 1 — Score: 4.633

operators → scada

Path 2 — Score: 3.351

operators → local_hmi

Path 3 — Score: 2.591

operators → local_hmi → industrial_network → plc_1 → sensors_actuators_1 → physical_process

Path 4 — Score: 2.591

operators → local_hmi → industrial_network → plc_2 → sensors_actuators_2 → physical_process

Path 5 — Score: 2.591

operators → scada → industrial_network → plc_1 → sensors_actuators_1 → physical_process

Methodology

This assessment uses a Bayesian network model constructed from the system topology. Each asset's base compromise probability is computed from its attributes (CVSS score, exposure, patch level). The Noisy-OR model generates conditional probability tables, and Variable Elimination performs inference given observed evidence. Risk scores combine posterior compromise probabilities with consequence impact scores.

Important: This report is a decision-support tool, not a guarantee of security. Risk scores are calculated approximations based on the configured model parameters and available evidence. They should be used alongside domain expertise and other security assessments.